Programming Fundamentals

Lab 3 Assignment

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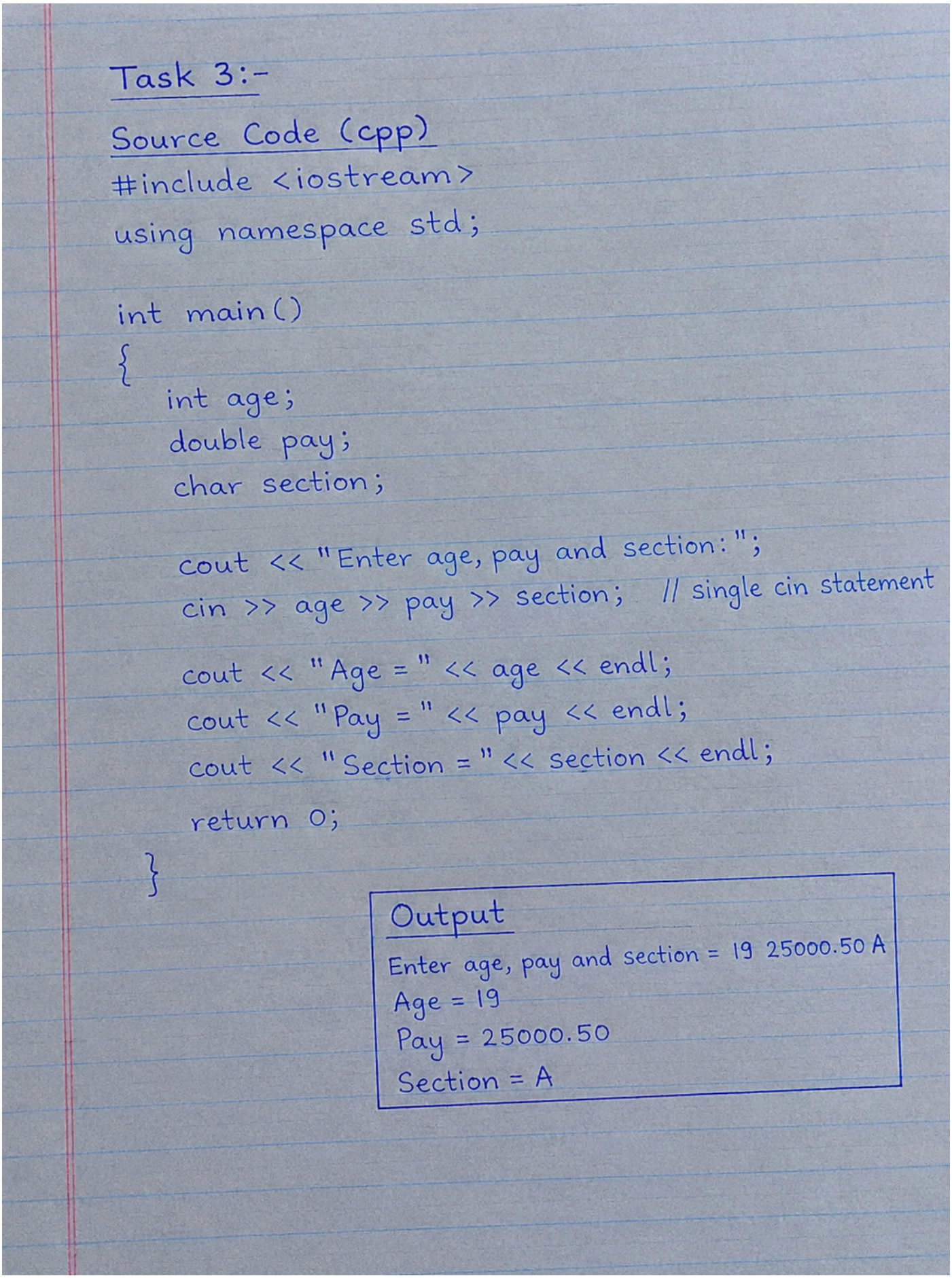
Roll No: 45

Class: BSCS

Section: A

Lab Task 1

Handwritten code



Task 3:-

Source Code (cpp)

```
#include <iostream>
using namespace std;

int main()
{
    int age;
    double pay;
    char section;

    cout << "Enter age, pay and section:";
    cin >> age >> pay >> section;   // single cin statement

    cout << "Age = " << age << endl;
    cout << "Pay = " << pay << endl;
    cout << "Section = " << section << endl;

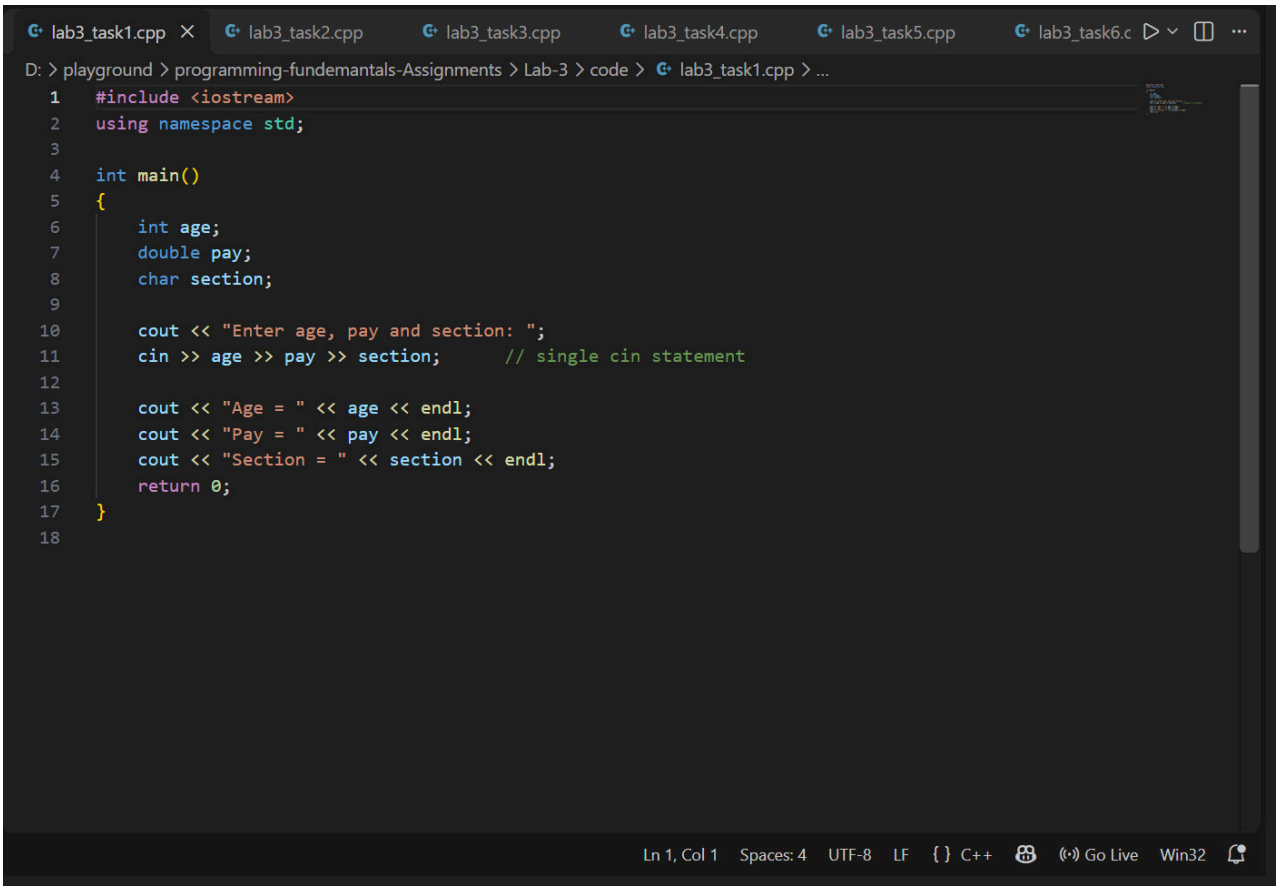
    return 0;
}
```

Output

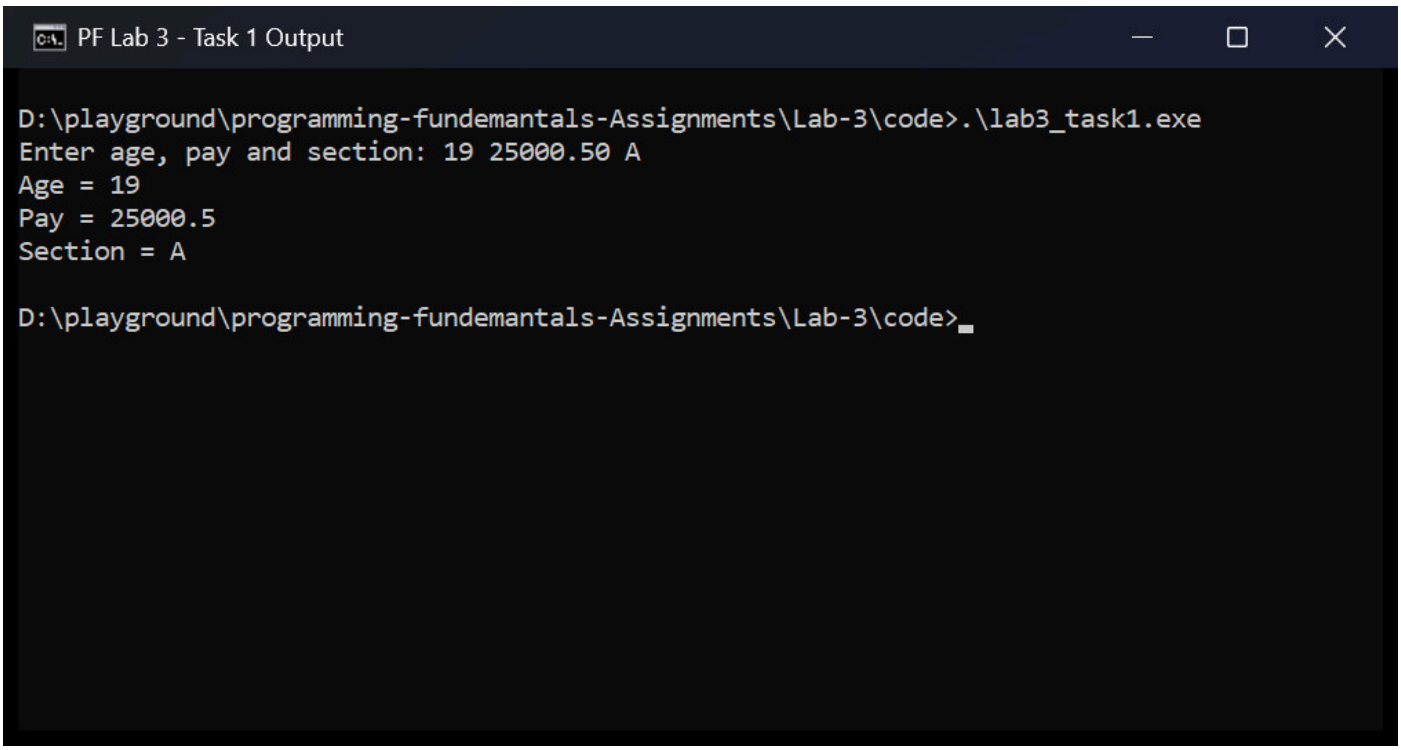
```
Enter age, pay and section = 19 25000.50 A
Age = 19
Pay = 25000.50
Section = A
```

Lab Task 1

Typed code (lab3_task1.cpp)

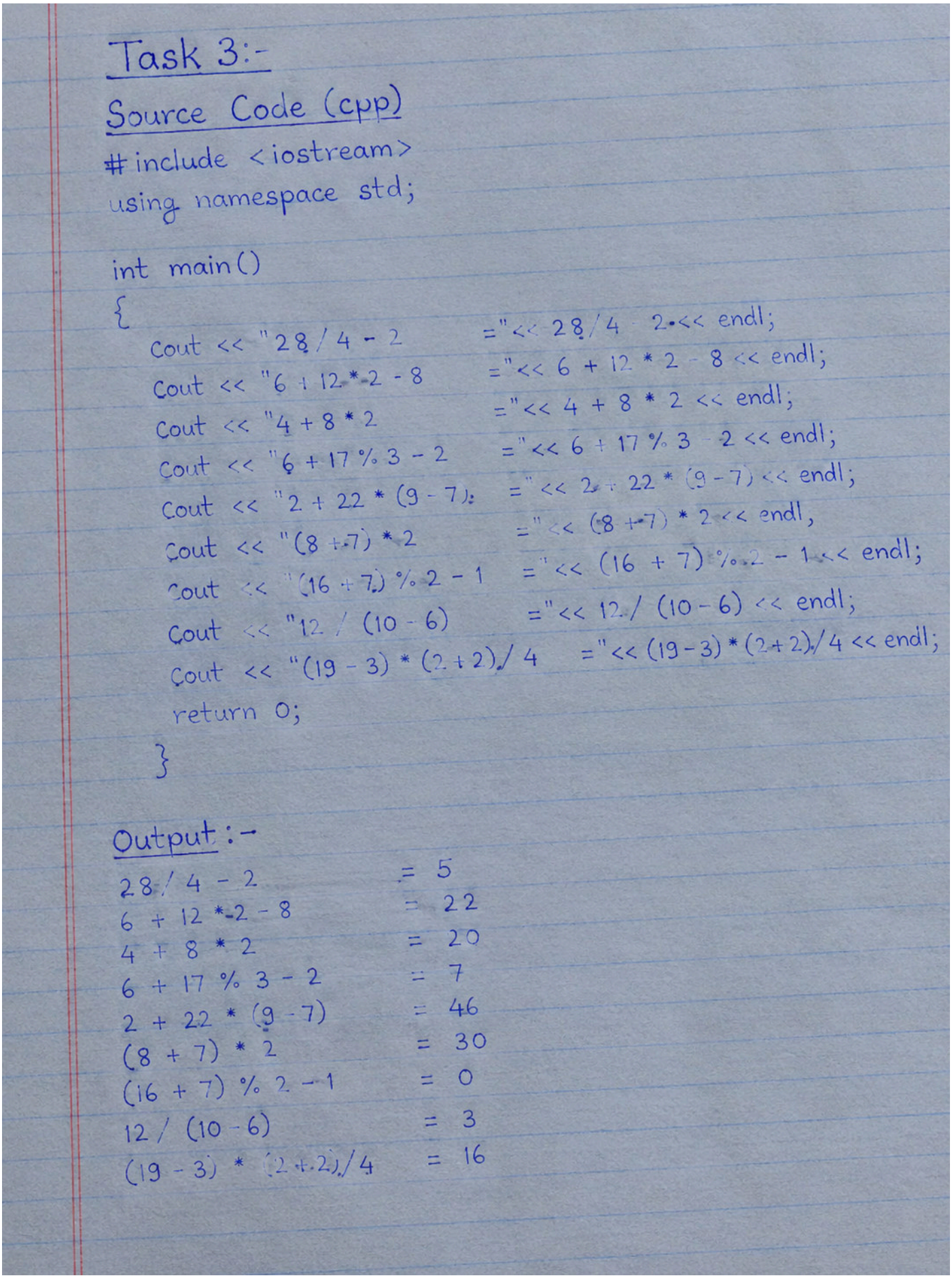


Output



Lab Task 2

Handwritten code



Task 3:-

Source Code (cpp)

```
#include <iostream>
using namespace std;

int main()
{
    Cout << "28 / 4 - 2          =" << 28 / 4 - 2 << endl;
    Cout << "6 + 12 * 2 - 8      =" << 6 + 12 * 2 - 8 << endl;
    Cout << "4 + 8 * 2           =" << 4 + 8 * 2 << endl;
    Cout << "6 + 17 % 3 - 2      =" << 6 + 17 % 3 - 2 << endl;
    Cout << "2 + 22 * (9 - 7);   =" << 2 + 22 * (9 - 7) << endl;
    Cout << "(8 + 7) * 2         =" << (8 + 7) * 2 << endl,
    Cout << "(16 + 7) % 2 - 1    =" << (16 + 7) % 2 - 1 << endl;
    Cout << "12 / (10 - 6)       =" << 12 / (10 - 6) << endl;
    Cout << "(19 - 3) * (2 + 2) / 4   =" << (19 - 3) * (2 + 2) / 4 << endl;
    return 0;
}
```

Output :-

```
28 / 4 - 2          = 5
6 + 12 * 2 - 8      = 22
4 + 8 * 2           = 20
6 + 17 % 3 - 2      = 7
2 + 22 * (9 - 7)    = 46
(8 + 7) * 2         = 30
(16 + 7) % 2 - 1    = 0
12 / (10 - 6)       = 3
(19 - 3) * (2 + 2) / 4   = 16
```

Lab Task 2

Typed code (lab3_task2.cpp)

lab3_task1.cpp | lab3_task2.cpp | lab3_task3.cpp | lab3_task4.cpp | lab3_task5.cpp | lab3_task6.c

D: > playground > programming-fundemantals-Assignments > Lab-3 > code > lab3_task2.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "28 / 4 - 2             = " << 28 / 4 - 2 << endl;
    cout << "6 + 12 * 2 - 8         = " << 6 + 12 * 2 - 8 << endl;
    cout << "4 + 8 * 2              = " << 4 + 8 * 2 << endl;
    cout << "6 + 17 % 3 - 2         = " << 6 + 17 % 3 - 2 << endl;
    cout << "2 + 22 * (9 - 7)       = " << 2 + 22 * (9 - 7) << endl;
    cout << "(8 + 7) * 2            = " << (8 + 7) * 2 << endl;
    cout << "(16 + 7) % 2 - 1       = " << (16 + 7) % 2 - 1 << endl;
    cout << "12 / (10 - 6)          = " << 12 / (10 - 6) << endl;
    cout << "(19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;
    return 0;
}
```

Ln 1, Col 2 Spaces: 4 UTF-8 LF C++ Go Live Win32

Output

PF Lab 3 - Task 2 Output

```
D:\playground\programming-fundemantals-Assignments\Lab-3\code>.\lab3_task2.exe
28 / 4 - 2             = 5
6 + 12 * 2 - 8         = 22
4 + 8 * 2              = 20
6 + 17 % 3 - 2         = 6
2 + 22 * (9 - 7)       = 46
(8 + 7) * 2            = 30
(16 + 7) % 2 - 1       = 0
12 / (10 - 6)          = 3
(19 - 3) * (2 + 2) / 4 = 16

D:\playground\programming-fundemantals-Assignments\Lab-3\code>
```

Lab Task 3

Handwritten code

Task 3:-

Source Code (cpp)

```
#include <iostream>
using namespace std;

int main()
{
   int units =5;
   float mass = 2.5;
   double weight;

   weight = mass * units;    // units is promoted to float, result is float

   cout << "Size of int    =" << sizeof(units) << " bytes" << endl;
   cout << "Size of float =" << sizeof(mass) << " bytes" << endl;
   cout << "Size of double =" << sizeof(weight) << " bytes" << endl;
   cout << "Size of (mass * units) =" << sizeof(mass * units) << "bytes (float)"
                                                              << endl;

   cout << "weight =" << weight << endl;
   cout << "Answer: B" << endl;

   return 0;
}
```

Output :

```
Size of int     = 4 bytes
Size of float   = 4 bytes
Size of double = 8 bytes
Size of (mass * units) =4 bytes (float)
weight = 12.5
Answer: B
```

Lab Task 3

Typed code (lab3_task3.cpp)

lab3_task1.cpp | lab3_task2.cpp | lab3_task3.cpp | lab3_task4.cpp | lab3_task5.cpp | lab3_task6.c

D: > playground > programming-fundemantals-Assignments > Lab-3 > code > lab3_task3.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main()
{
    int units = 5;
    float mass = 2.5;
    double weight;

    weight = mass * units;   // units is promoted to float, result is float

    cout << "Size of int    = " << sizeof(units) << " bytes" << endl;
    cout << "Size of float  = " << sizeof(mass) << " bytes" << endl;
    cout << "Size of double = " << sizeof(weight) << " bytes" << endl;
    cout << "Size of (mass * units) = " << sizeof(mass * units) << " bytes (float)" << endl;
    cout << "weight = " << weight << endl;
    cout << "Answer: B" << endl;
    return 0;
}
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF C++ Go Live Win32

Output

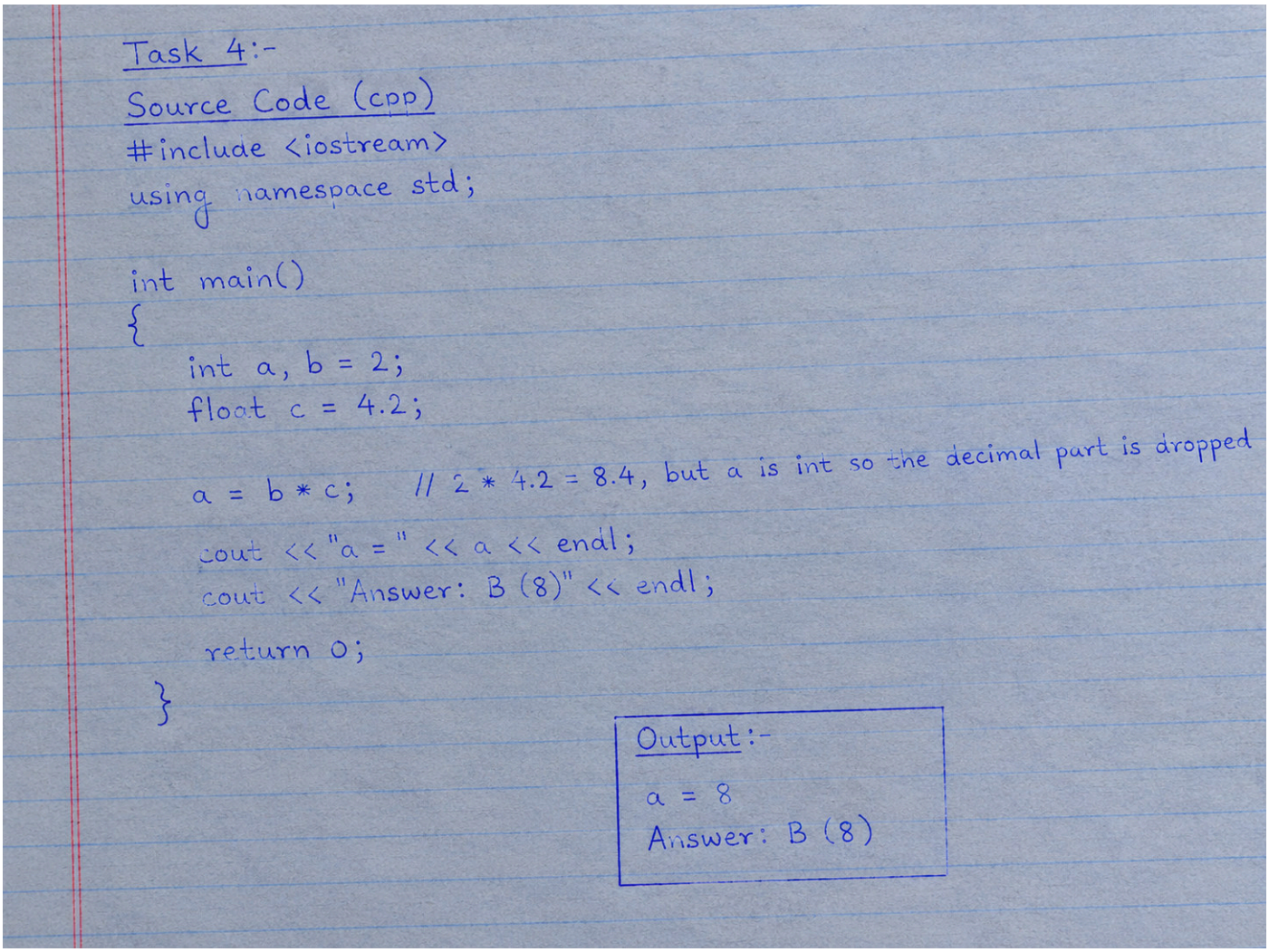
PF Lab 3 - Task 3 Output

```
D:\playground\programming-fundemantals-Assignments\Lab-3\code>.\lab3_task3.exe
Size of int    = 4 bytes
Size of float  = 4 bytes
Size of double = 8 bytes
Size of (mass * units) = 4 bytes (float)
weight = 12.5
Answer: B

D:\playground\programming-fundemantals-Assignments\Lab-3\code>
```

Lab Task 4

Handwritten code



Task 4:-

Source Code (cpp)

```
#include <iostream>
using namespace std;

int main()
{
    int a, b = 2;
    float c = 4.2;

    a = b * c;    // 2 * 4.2 = 8.4, but a is int so the decimal part is dropped

    cout << "a = " << a << endl;
    cout << "Answer: B (8)" << endl;

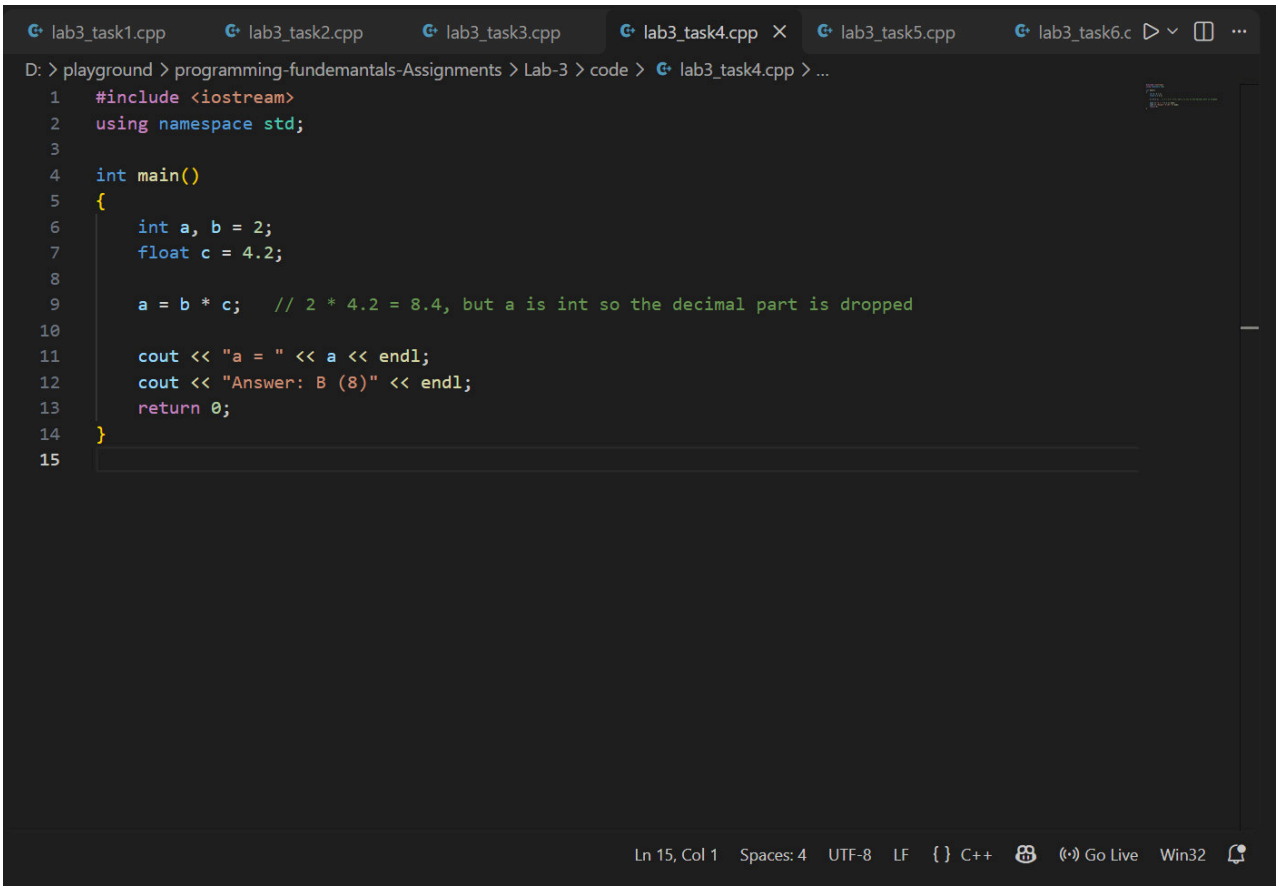
    return 0;
}
```

Output:-

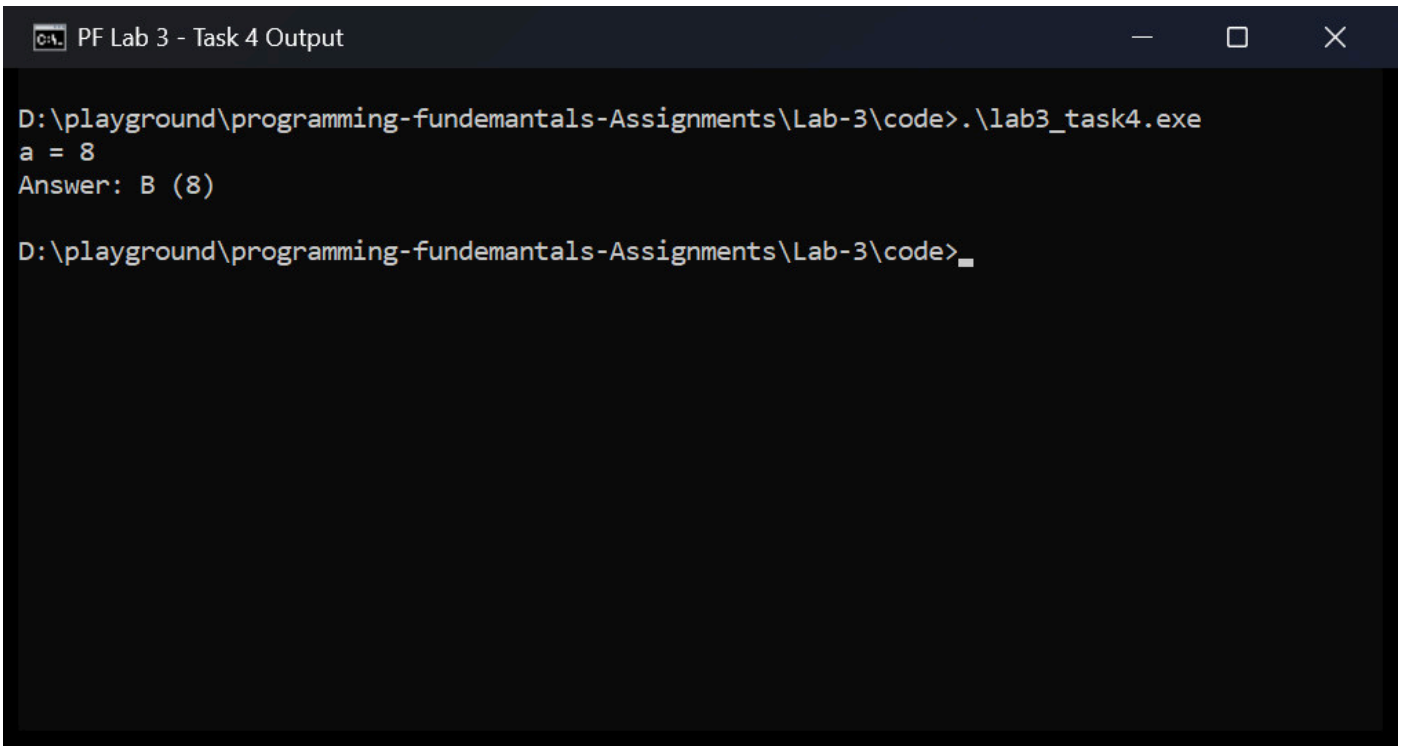
```
a = 8
Answer: B (8)
```

Lab Task 4

Typed code (lab3_task4.cpp)



Output



Lab Task 5

Handwritten code

Task 5:-

Source Code (cpp)

```cpp
#include <iostream>
using namespace std;

int main()
{
    int qty = 10, salesReps = 4;
    double unitsEach;

    unitsEach = qty / salesReps;          // integer division
    cout << "Without type cast: " << unitsEach << endl;

    unitsEach = static_cast<double>(qty) / salesReps;  // type cast
    cout << "With type cast: " << unitsEach << endl;

    return 0;
}
```

Output:-

```
Without type cast: 2
With type cast: 2.5
```

Lab Task 5

Typed code (lab3_task5.cpp)

lab3_task1.cpp | lab3_task2.cpp | lab3_task3.cpp | lab3_task4.cpp | lab3_task5.cpp | lab3_task6.c

D: > playground > programming-fundemantals-Assignments > Lab-3 > code > lab3_task5.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main()
{
    int qty = 10, salesReps = 4;
    double unitsEach;

    unitsEach = qty / salesReps;                    // integer division
    cout << "Without type cast: " << unitsEach << endl;

    unitsEach = static_cast<double>(qty) / salesReps;  // type cast
    cout << "With type cast: " << unitsEach << endl;
    return 0;
}

```

Ln 1, Col 1 Spaces: 4 UTF-8 LF { } C++ Go Live Win32

Output

PF Lab 3 - Task 5 Output

```
D:\playground\programming-fundemantals-Assignments\Lab-3\code>.\lab3_task5.exe
Without type cast: 2
With type cast: 2.5

D:\playground\programming-fundemantals-Assignments\Lab-3\code>
```

Lab Task 6

Handwritten code

Task 6:

Source Code (cpp):

```cpp
#include <iostream>
#include <iomanip>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(time(0));

    int num1 = rand() % 900 + 100; // random number from 100 to 999
    int num2 = rand() % 900 + 100;

    cout << " " << setw(4) << num1 << endl;
    cout << "+" << setw(4) << num2 << endl;
    cout << "----" << endl;

    system("pause");

    cout << " " << setw(4) << num1 << endl;
    cout << "+" << setw(4) << num2 << endl;
    cout << "----" << endl;
    cout << " " << setw(4) << num1 + num2 << endl;

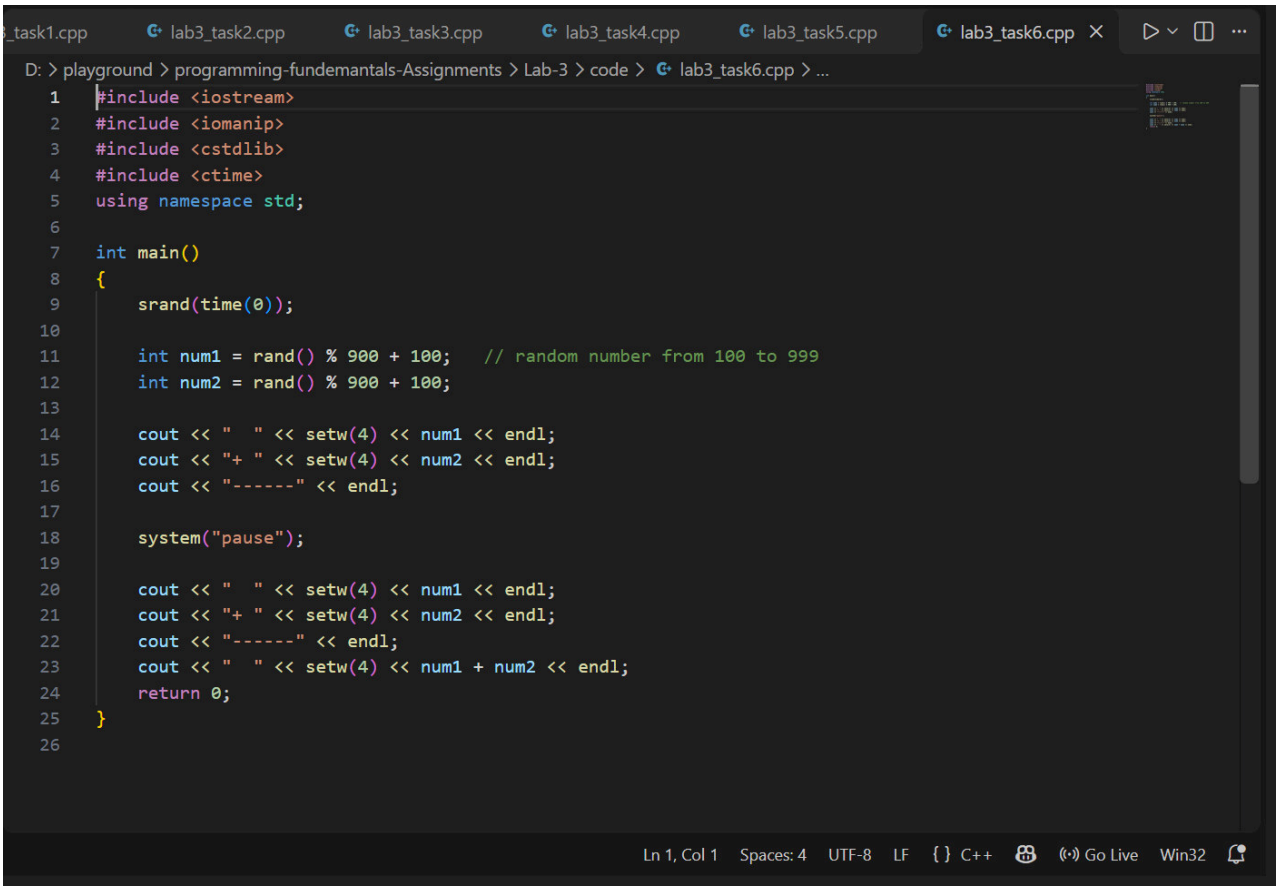
    return 0;
}
```

Output:

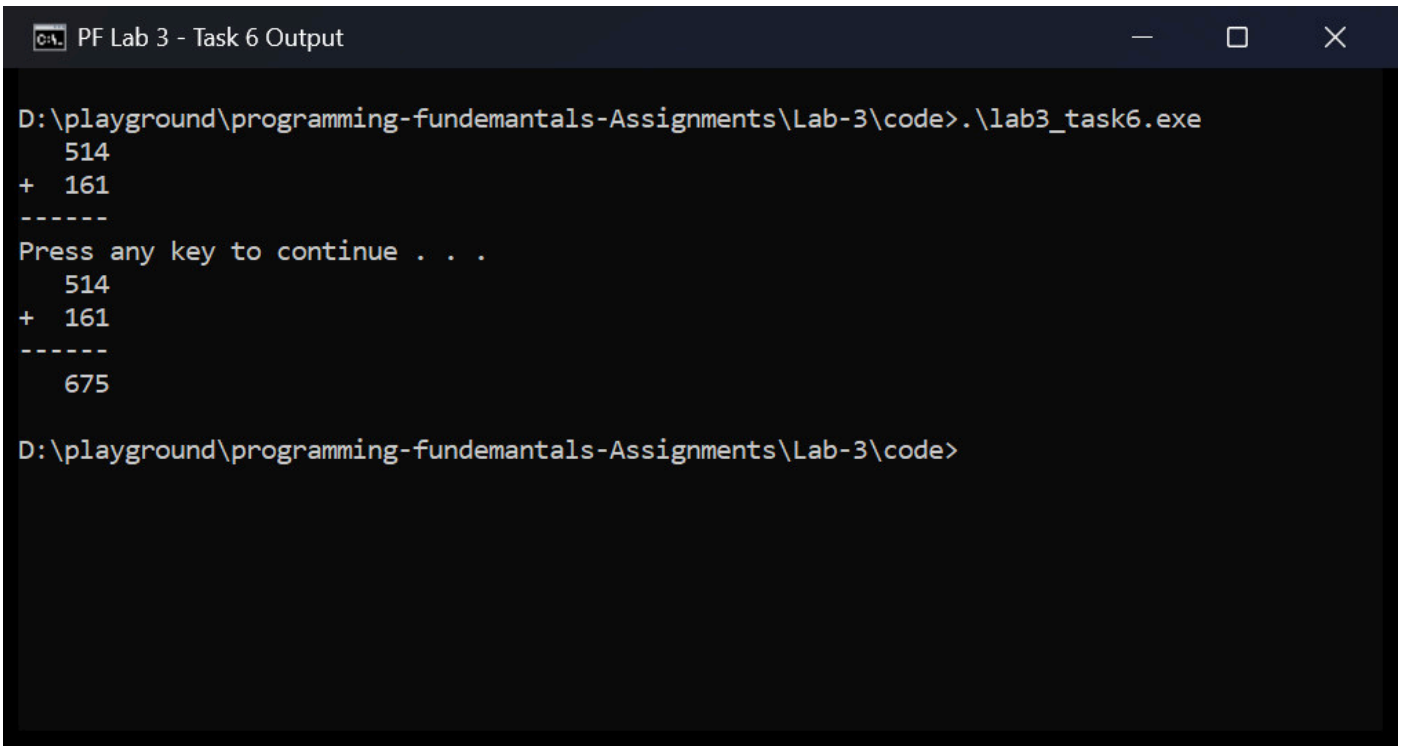
```
 514
+161
-----
 675
```

Lab Task 6

Typed code (lab3_task6.cpp)



Output



Home Task 1 - Program 1

Handwritten code

Task 7:

Source Code (cpp):

```
#include <iostream>    // error 1: <iostream> was missing
using namespace std;

int main(){
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;    // error 3: cin should be cin, error 4: << should be >>
    cout << "Enter another number: ";
    cin >> number2;    // error 5: << should be >>

    sum = number1 + number2;    // error 6: number1 + number2 = sum was reversed

    cout << "The sum of the two numbers is " << sum;    // error 7, 8: << and ; were missing

    return 0;

}
```

Output:

```
Enter a number: 4.5
Enter another number: 3
The sum of the two numbers is 7.5
```

Output:

Enter a number: 4.5
Enter another number: 3
The sum of the two numbers is 7.5

Home Task 1 - Program 1

Typed code (lab3_home_task1_program1.cpp)

lab3_task3.cpp | lab3_task4.cpp | lab3_task5.cpp | lab3_task6.cpp | lab3_home_task1_program1.cpp

D: > playground > programming-fundemantals-Assignments > Lab-3 > code > lab3_home_task1_program1.cpp > ...

```cpp
#include <iostream>       // error 1: #include <iostream> was missing
using namespace std;

int main()                // error 2: void main() should be int main()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;       // error 3: Cin should be cin, error 4: << should be >>
    cout << "Enter another number: ";
    cin >> number2;       // error 5: << should be >>
    sum = number1 + number2;   // error 6: number1 + number2 = sum was reversed
    cout << "The sum of the two numbers is " << sum;   // error 7, 8: << and ; were missing
    return 0;
}

```

Ln 1, Col 1 Spaces: 4 UTF-8 LF { } C++ Go Live Win32

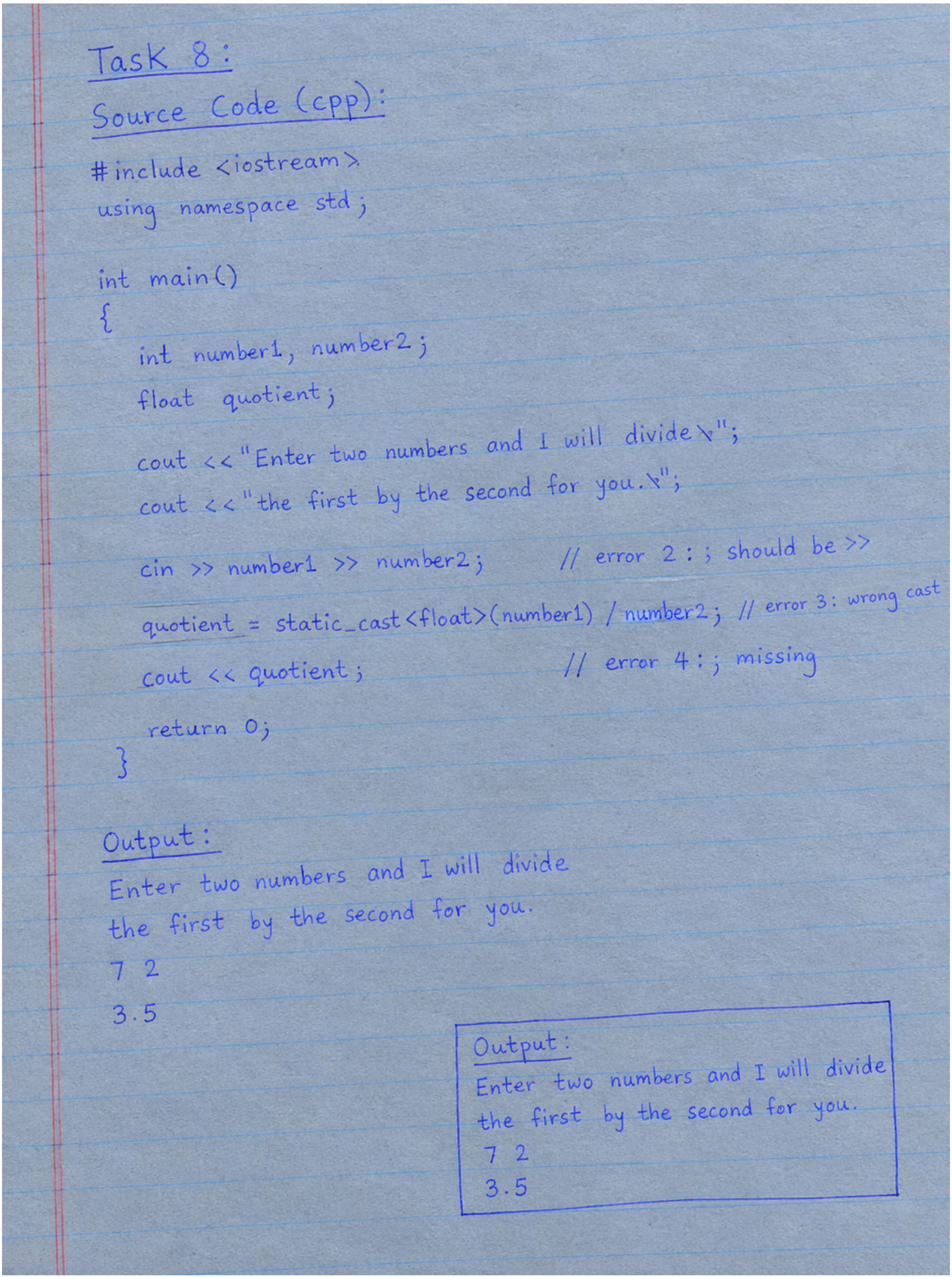
Output

PF Lab 3 - Home Task 1 Program 1 Output

```
D:\playground\programming-fundemantals-Assignments\Lab-3\code>.\lab3_home_task1_program1.e
xe
Enter a number: 4.5
Enter another number: 3
The sum of the two numbers is 7.5
D:\playground\programming-fundemantals-Assignments\Lab-3\code>
```

Home Task 1 - Program 2

Handwritten code



Task 8:

Source Code (cpp):

```
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";

    cin >> number1 >> number2;          // error 2 : ; should be >>
    quotient = static_cast<float>(number1) / number2; // error 3: wrong cast
    cout << quotient;                   // error 4 : ; missing

    return 0;
}
```

Output:

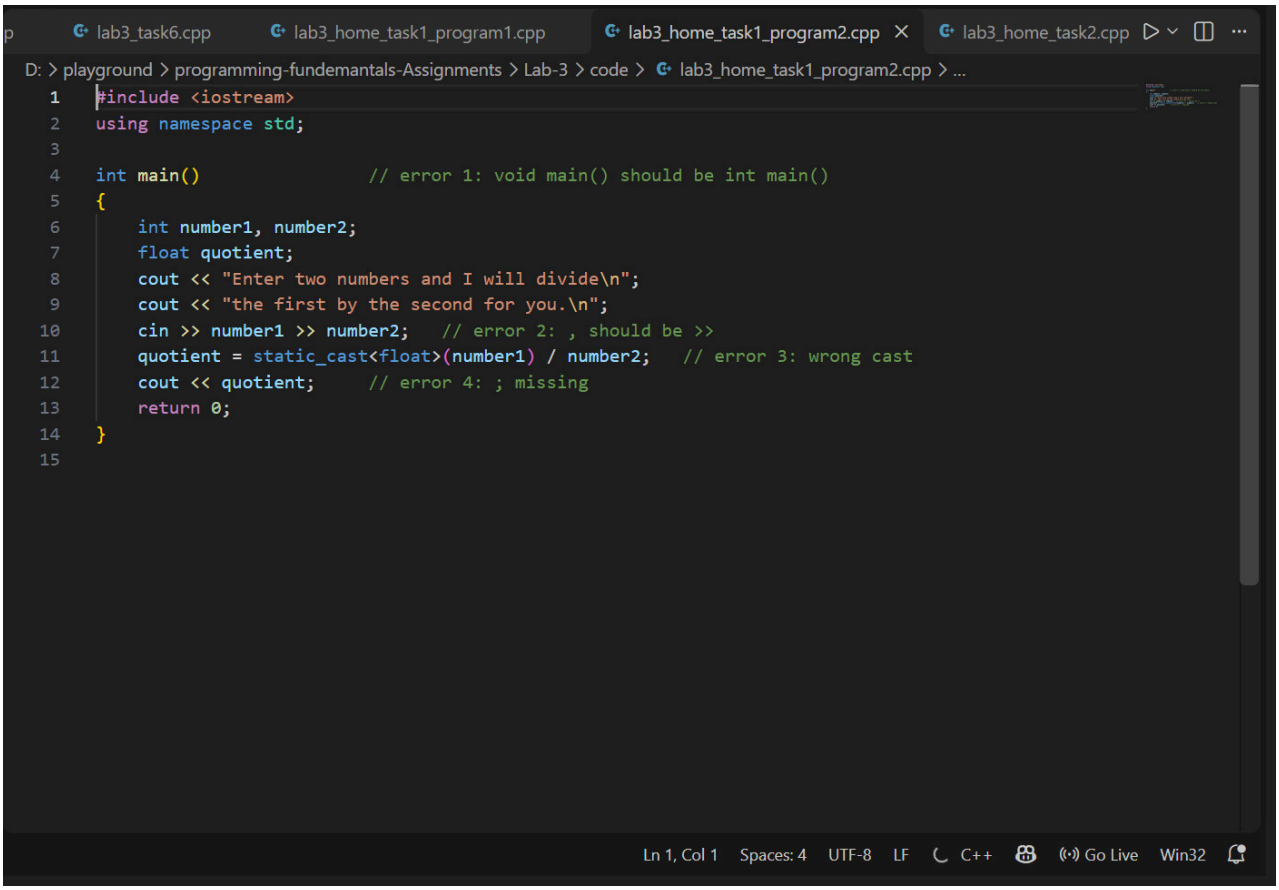
```
Enter two numbers and I will divide
the first by the second for you.
7 2
3.5
```

Output:

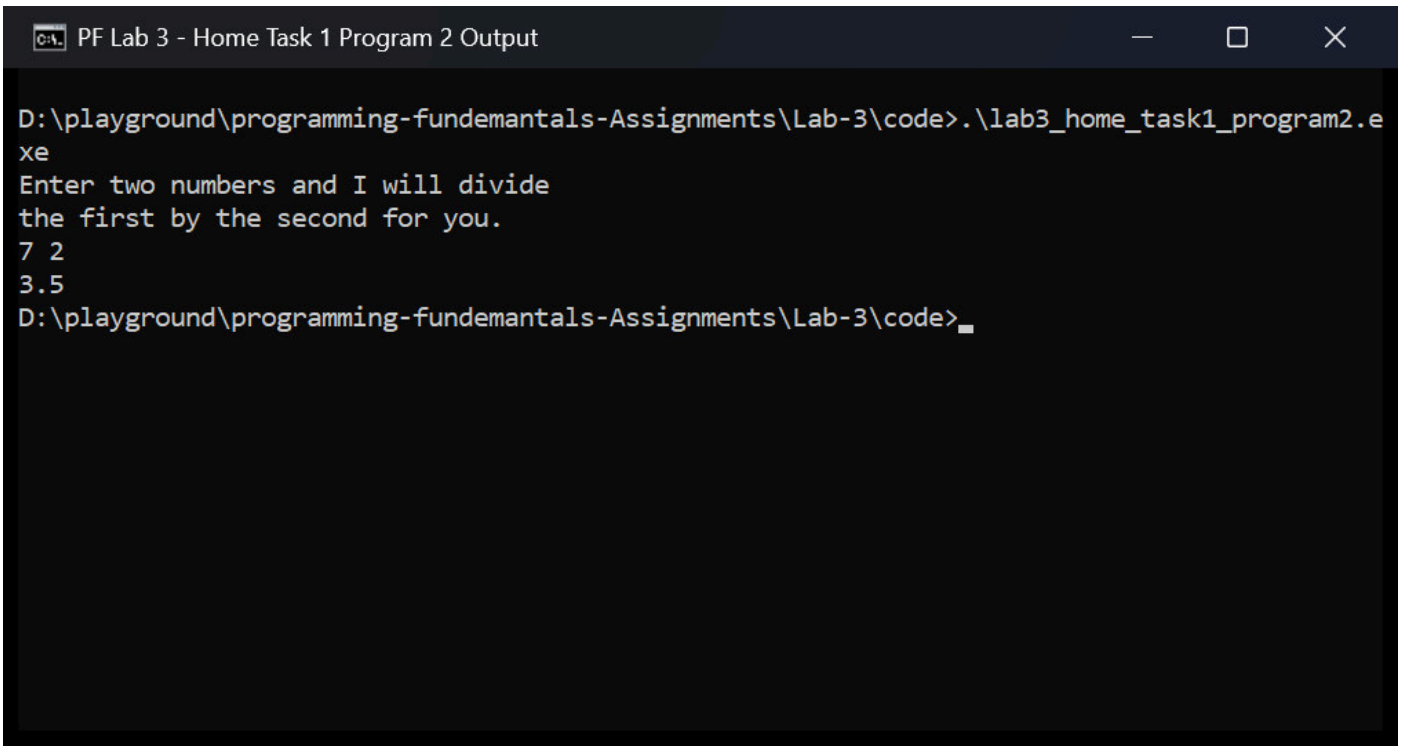
Enter two numbers and I will divide
the first by the second for you.
7 2
3.5

Home Task 1 - Program 2

Typed code (lab3_home_task1_program2.cpp)

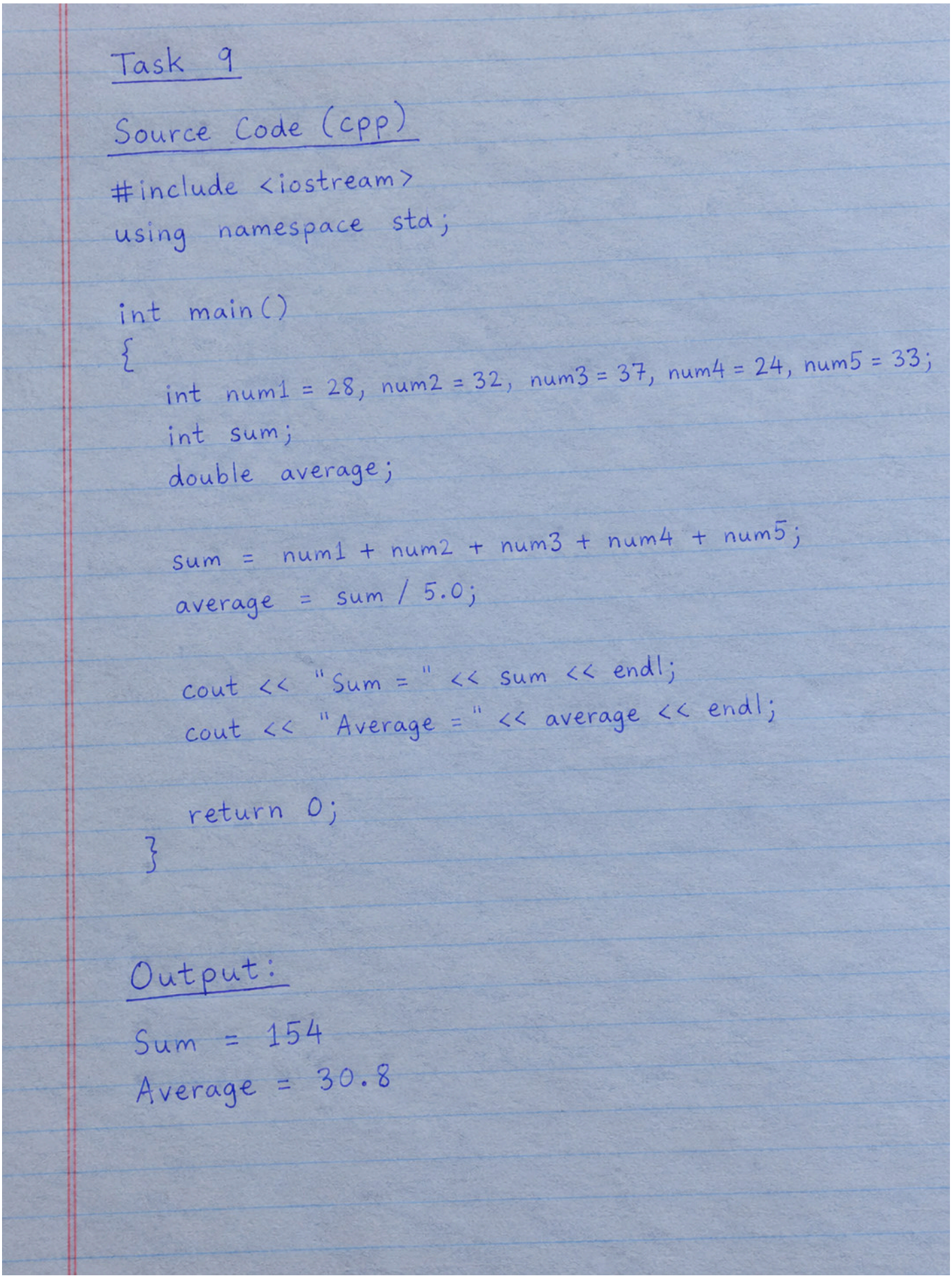


Output



Home Task 2

Handwritten code



Task 9

Source Code (cpp)

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1 = 28, num2 = 32, num3 = 37, num4 = 24, num5 = 33;
    int sum;
    double average;

    sum = num1 + num2 + num3 + num4 + num5;
    average = sum / 5.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

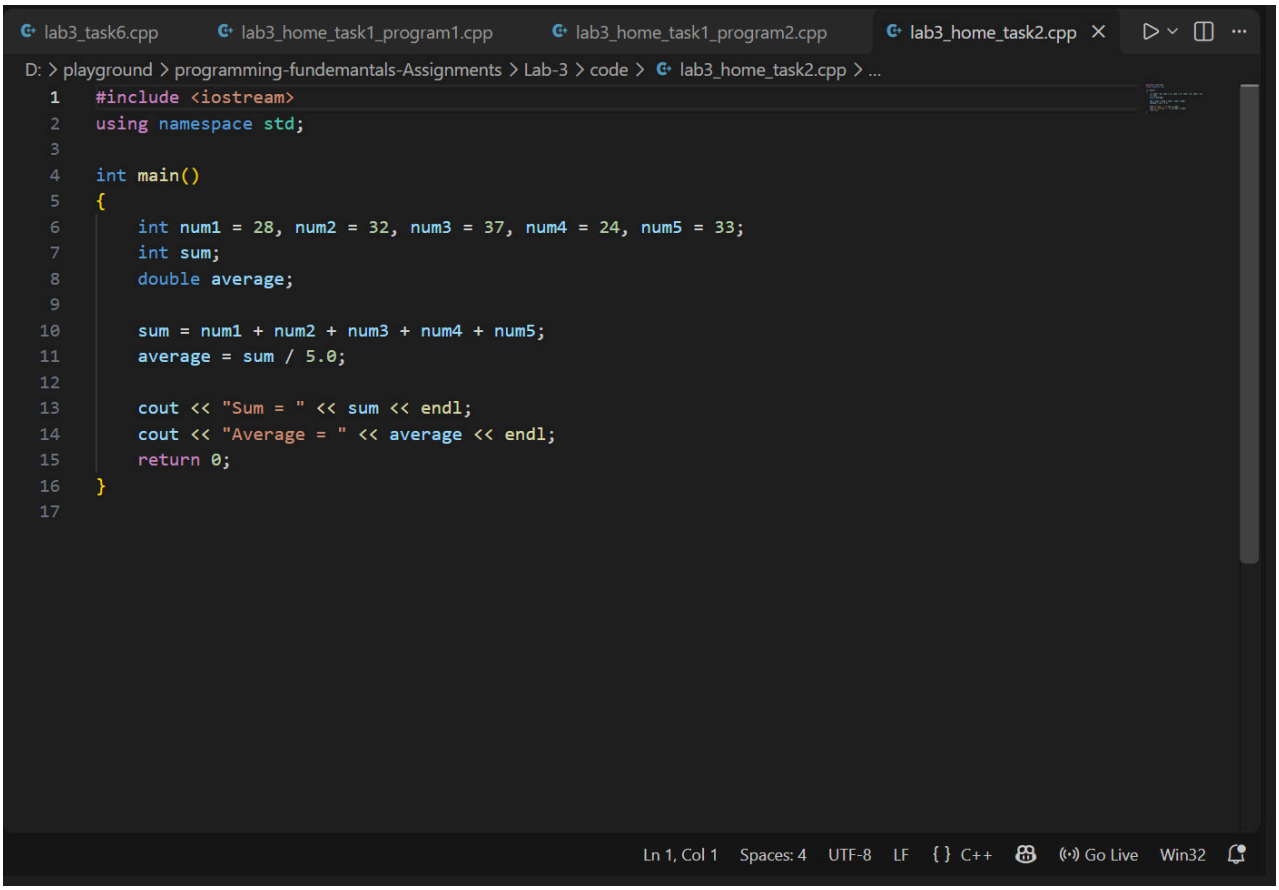
    return 0;
}
```

Output:

```
Sum = 154
Average = 30.8
```

Home Task 2

Typed code (lab3_home_task2.cpp)



Output

